

When Do Audited Action Gates Help? The Cost and Failure Conditions of Epistemic Strictness in LLM Agents

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Abstract

Recent work equips LLM agents with mechanisms that verify internal beliefs before committing to actions, and reports that such mechanisms improve faithfulness. We ask the complementary question: *what do these mechanisms cost, and when do they fail?* We build a synthetic fault-diagnosis environment in which genuine causal rules and constructed spurious associations are indistinguishable from observation and separable only by intervention, so that epistemic strictness carries a real price. Across 1,330 completed agent runs on nine models we report four findings. (i) Increasing strictness suppresses superstition and discovery together: spurious-rule adoption falls by 1.51 of 4 and true-rule recall by 0.111 (paired, $n=57$, Holm-corrected $p < 0.02$), but the size of the discovery cost varies sharply across models, so there is no single model-independent frontier. (ii) Self-reported epistemic status is unreliable: agents label rules VERIFIED whose causal component was never intervened upon, and a gate that trusts those labels yields no detectable benefit, whereas a harness-audited gate does. (iii) Restricting action to audited rules improves downstream diagnosis by +0.114 [+0.019, +0.208] ($n=55$ paired, four models, held-out evaluation half), but the benefit is concentrated almost entirely in systems that were already performing poorly: +0.31 when the ungated baseline is below 0.60 versus +0.01 at 0.80–0.95 (independent-covariate design, $r = -0.537$, permutation $p=0.0007$). (iv) The gate’s effect *changes sign* with the verification budget: with no budget it *reduces* accuracy by 0.315, and the budget-by-gate interaction (DiD = +0.456 [+0.297, +0.625]) is the largest effect we measure. We also refute our own pre-registered hypothesis that over-strict agents stop exploring: intervention usage is flat across six strictness levels. An audited action gate does not make an agent smarter; it prevents a belief set already contaminated by coincidences from degrading action—and only when the system can afford to verify.

1 Introduction

A recurring assumption in work on LLM agents is that more caution is better: an agent that abstains under uncertainty, or that verifies its beliefs before acting, is treated as strictly improved. The tendency to propose causal hypotheses on limited evidence, however, drives two things at once: the adoption of coincidences (false positives) and the discovery of genuine but not-yet-confirmed rules (true positives). If both are driven by the same disposition, suppressing one should suppress the other.

The abstention literature has quantified a related trade-off in question answering, where over-abstention reduces usefulness on questions the model could have answered. That is not the trade-off studied here. Our cost lands on a different quantity: the *discovery of new causal rules* in an interactive setting where the agent must decide whether to spend a budget on experiments.

Why a synthetic environment. Testing this requires an environment in which (a) genuine rules and spurious associations are indistinguishable in observational data, (b) only intervention separates them, and (c) ground truth is known by construction so that no human annotation is needed. Natural corpora do not satisfy all three.

Pre-registered hypotheses. We fixed three hypotheses before collecting data. **T1** (trade-off): raising strictness lowers both spurious adoption and true-rule discovery. **T2** (mechanism): separating *holding* a hypothesis from *acting* on it pushes the frontier outward. **T3** (non-monotonicity): an over-strict agent stops exploring and discovery collapses. Section 5 reports which survived; Appendix A lists every hypothesis of our own that the data refuted.

Contributions. We do not propose a new mechanism. Verification-before-commitment and auditable hypothesis protocols already exist [Yuan et al. \(2026\)](#); [HEP \(2026\)](#), as do synthetic interactive causal-discovery environments [CausaLab \(2026\)](#); [Interventional Agents \(2026\)](#). Existing work reports that such mechanisms work. We measure what they cost and when they stop working:

1. a quantified discovery cost of epistemic strictness, and evidence that its magnitude is model-dependent rather than universal (§5.2);
2. evidence that model-*self-reported* epistemic status is systematically unfaithful, so gates built on self-labels are ineffective, while gates built on harness-side interventional audit are not (§5.3);
3. the moderation structure of the gate’s benefit, measured with a split-half design that removes the shared-noise artefact (§5.4);
4. a sign-flipping budget-by-gate interaction: without verification budget, the gate is harmful (§5.5);
5. full release of code, environment, per-run checkpoints and prompts.

2 Related work

Abstention. [Liu et al. \(2026\)](#) build a paired-task benchmark for agentic abstention across 42 sandbox environments and 17 frontier models, and report that abstention capability is largely independent of general task-solving capability. [Luo et al. \(2026\)](#) treat abstention as a sequential decision and evaluate 13 agent systems on more than 28,000 tasks. These works measure whether an agent abstains when it should; we measure what strictness costs on the discovery side.

Verification before action. [Yuan et al. \(2026\)](#) propose SAVeR, which enforces verification over internal belief states before action commitment via adversarial self-auditing, and report improved faithfulness *while preserving competitive end-task performance*. The Hypothesis Evolution Protocol ([HEP, 2026](#)) makes hypothesis generation, evaluation and evolution explicit and auditable, and reports that stronger base models exploit the protocol more fully. Our audit differs in substrate:

Table 1: Environment invariant (20,000 samples, base configuration).

Relation	Observational $P(s \mid c=1)$	Interventional $P(s \mid do(c=1))$
True rules ($K=3$)	0.902, 0.904, 0.897	0.902, 0.903, 0.899
Spurious pairs ($M=4$)	0.773, 0.740, 0.777, 0.740	0.503, 0.496, 0.497, 0.502

SAVeR checks internal logical and evidential consistency using the model itself, whereas our gate checks each rule against *external interventional records* produced by the harness, which the model cannot fabricate. Our data bear directly on the self-auditing assumption (§5.3), and our moderation result stands in tension with that capability finding (§6).

Interactive causal discovery. CausaLab (CausaLab, 2026) places agents in a synthetic laboratory with prior records and budgeted interventions, and reports a persistent gap between task accuracy and mechanism recovery. Related work argues that progress should be measured by interventional outcomes against a fixed hidden structural causal model rather than by the agent’s own narrative (Interventional Agents, 2026). Our environment shares the budgeted-intervention setting; it differs in that the confounding structure is *fixed by construction*, so a specific set of M spurious pairs is known in advance and “spurious-rule adoption” can be used directly as a dependent variable.

Memory and propagation. Tang et al. (2026) show that spurious correlations persisted in agentic memory become self-reinforcing, because the actions they induce generate the next trajectory written back to memory. Margalit et al. (2026) formalise shared-memory failure modes including provenance collapse. These lines concern propagation across steps and agents; the present paper concerns the genesis of spurious belief within a single agent.

3 The Fault World environment

3.1 Structure

A synthetic machine is described by components and symptoms named with random semantics-free symbols, re-randomised for every run to exclude memorisation. The base configuration has 6 components and 8 symptoms, giving a hypothesis space of 48 candidate rules:

- $K=3$ *true causes*: component c_i causes symptom s_i with $P = 0.90$; baseline occurrence without the cause is 0.05.
- 2 *confounded components*: each is driven by a hidden latent Z that also drives two symptoms, producing $M=4$ *spurious pairs*.
- one pure-noise component and one pure-noise symptom.

3.2 The invariant that makes strictness costly

Table 1 reports 20,000-sample checks. Observationally, spurious pairs look like moderately strong rules; under intervention they collapse to the 0.5 base rate while true rules are unchanged. An agent therefore cannot separate them without spending intervention budget.

3.3 Parameterised variants

For robustness we vary the generative parameters: WEAK (observational co-occurrence 0.65/0.62), STRONG (0.86/0.83), RICH ($K=4$, $M=2$) and POOR ($K=2$, $M=6$). The interventional invariant holds in all variants.

3.4 Agent loop

Each run consists of three episodes. In each episode the agent receives 15 fresh observation logs plus the results of its previous interventions, and returns its *complete current belief set*—each rule tagged HYPOTHESIS or VERIFIED with a confidence—together with the interventions it wishes to run. Each intervention forces one component active or inactive and runs the machine 20 times, returning symptom counts. A downstream diagnostic task follows.

4 Experimental design

4.1 Independent variables

S1, prompt-level strictness, six levels: L0 record anything remotely possible; L1 record on plausible indication; L2 record when evidence is reasonably consistent; L3 record only when confident it is genuinely causal; L4 record only after an intervention has tested it; L5 record only after testing the component in both directions with a clear difference. **S2, intervention budget**: 0/3/6/12/21 per run. **S4, action gate**: OFF (diagnosis may use the entire belief set), SELF (only rules the model labelled VERIFIED), AUDIT (only rules that pass the harness audit).

4.2 Harness audit criterion

A rule $c \rightarrow e$ passes iff the run’s intervention log contains either $do(c=ACTIVE)$ with e occurring $\geq 14/20$, or $do(c=INACTIVE)$ with e occurring $\leq 6/20$. Each threshold has one-sided false-positive rate $P(X \geq 14 \mid p=0.5) = 0.058$ against the spurious null; true rules generate $p=0.90$. The model’s self-reported status is treated as a measured object, never as gate input.

4.3 Downstream task and split-half measurement

Each diagnostic case activates exactly one true-cause component and asks which component to repair. We use 30 cases per run, of which typically 23–26 are uniquely answerable by an oracle holding the true rule set (a true effect fails to appear with probability 0.10, and two true effects sometimes co-occur; including unanswerable cases only dilutes signal—the oracle ceiling over all cases is 0.81).

Answerable cases are split by index parity into two disjoint halves: **A** (even), used only to produce an *independent* baseline covariate, and **B** (odd), on which all effects are reported. This is necessary: the gate effect $\Delta = \text{AUDIT} - \text{OFF}$ shares noise with its own baseline OFF, so without the split a moderation analysis cannot be distinguished from regression to the mean.

4.4 Statistics

All conditions share the same world seeds, so comparisons are paired. We report bootstrap 95% confidence intervals (10,000 resamples), p -values from paired sign-flip permutation tests, and Holm–Bonferroni correction within families of contrasts. All contrasts are reported, including non-significant ones.

Table 2: Strictness LOW→HIGH, three models, $n=57$ paired runs.

Measure	Δ	p_{Holm}	Verdict
Spurious rules adopted (of 4)	−1.509	.0023	significant
True-rule recall	−0.111	.0117	significant
Interventions used	−0.526	.108	not significant
Audited true rules	−0.029	.661	not significant

4.5 Models and fixed parameters

Nine models were used across experiments: `qwen3.8-27b`, `glm-4.5-air`, `glm-5.2`, `kimi-k3`, `qwen3.7-flash-2026-06-08`, `qwen3.7-max-2026-06-08`, `qwen3.7-plus`, `qwen3.7-plus-2026-05-26`, `deepseek-v4-flash-0731`. The model set differs between experiments because of free API quota availability, so results are not pooled across experiments. Fixed parameters: chain-of-thought disabled (`enable_thinking=false`), 15 observations per episode, 3 episodes, 20 trials per intervention, temperature 1.0. Disabling chain-of-thought was necessary for feasibility (each call otherwise took 60–150 s) and is an untested simplification; see §7.

5 Results

5.1 The environment induces superstition

Spurious-rule adoption falls inside the pre-registered 0.15–0.85 band (belief-level mean ≈ 0.71 at full scale). Purely observational conditions reach 0.90, but that is *not* superstition: adopting an association when only observational data exist is rational. The main experiments are therefore run under conditions where intervention is available.

Illustratively, in one run the agent intervened on a confounded component and observed the two spurious symptoms at 6/20 and 7/20 (versus $\approx 15/20$ observationally), yet still recorded both spurious rules at confidence 0.7–0.85, reasoning that the intervention results were “consistent with the primary associations” and attributing the deviation to randomness.

5.2 T1: the trade-off exists, but there is no common frontier

Strictness reliably suppresses superstition and also costs recall (Table 2). The magnitude of the cost, however, is strongly model-dependent: recall $\Delta = -0.283$ $[-0.417, -0.150]$ for one model, versus -0.017 and -0.020 (both unstable) for the other two. The strong form of T1—that all strictness settings lie on one FDR–recall frontier—is *not* supported: some models suppress superstition almost for free.

5.3 Self-reported epistemic status is unreliable

In one model, the high-strictness condition produced on average 1.33 rules labelled VERIFIED whose causal component had never been intervened upon at all. The effect did not replicate in two other models ($\Delta = +0.15$ and -0.20 , both with CIs spanning zero), so we report it as a failure mode rather than a general claim.

Its methodological consequence is firm. Under the earlier 8-case protocol, `OFF` → `SELF` gave $+0.090$ and `SELF` → `AUDIT` gave $+0.087$; neither survived Holm correction ($p_{\text{Holm}} = .204$), while

Table 3: Gate effect (AUDIT-OFF) on the held-out evaluation half, 30-case protocol.

Model	Δ	n
deepseek-v4-flash-0731	+0.130 [−0.015, +0.260]	20
glm-5.2	+0.142 [−0.054, +0.327]	20
qwen3.7-flash-2026-07-15	+0.065	9
qwen3.7-plus-2026-05-26	+0.044	6
Pooled	+0.114 [+0.019, +0.208]	55 ($p=0.024$)

Table 4: Budget $\times$ gate, two models, 20 repetitions, 8-case protocol.

Contrast	Δ	p_{Holm}
Gate effect at budget 0	−0.315 [−0.452, −0.178]	.0017
Gate effect at budget 6	+0.140 [+0.054, +0.227]	.0080
Gate effect at budget 21	+0.081 [+0.002, +0.163]	.063
Interaction (DiD), budget 0 $\rightarrow$ 6	+0.456 [+0.297, +0.625]	.0017
Interaction (DiD), budget 0 $\rightarrow$ 21	+0.320 [+0.165, +0.477]	.0017

the combined OFF $\rightarrow$ AUDIT did. Trusting self-reported labels captures at best half of the available benefit.

5.4 T2: the gate works, but its benefit is strongly moderated

No single model survives Holm correction (best $p_{\text{Holm}} = 0.36$); the effect is detectable only pooled (Table 3).

The moderation is measured with the independent covariate: the x -axis is OFF accuracy on development half A, the y -axis is the gate effect on evaluation half B, and the two share no cases. We find $r = -0.537$, $n = 55$, permutation $p = 0.0007$:

Ungated baseline (half A)	n	Gate effect (half B)
< 0.60	19	+0.308 [+0.170, +0.440]
0.60–0.80	15	+0.103 [−0.054, +0.234]
0.80–0.95	11	+0.006 [−0.243, +0.216]
≥ 0.95	10	−0.117 [−0.301, +0.036]

The top bin is subject to a structural ceiling—when the baseline approaches 1.0, Δ can only be non-positive—and must not be read as “the gate is harmful”. The informative range is below 0.80, where the benefit decays monotonically from +0.31 to +0.01.

Crucially, when the gate helps, *the belief set is not cleaner*: in the highest-benefit conditions the agent still holds 2.5–3.3 spurious rules of 4. The gain comes from acting more strictly, not from believing less.

5.5 Budget by gate: without verification budget the gate is harmful

With zero budget the audited set is necessarily empty, so the gate strips the agent of every usable rule; one model falls to 0.28 accuracy. The interaction (Table 4, Figure 1) is the largest effect we

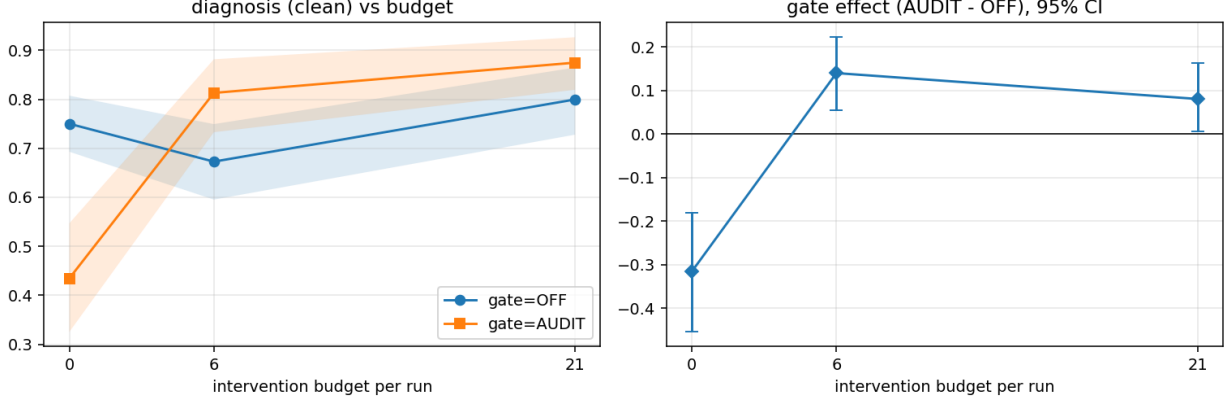


Figure 1: Left: downstream accuracy under both gate settings as a function of intervention budget. Right: the gate effect with 95% CIs; the sign flips between budget 0 and 6.

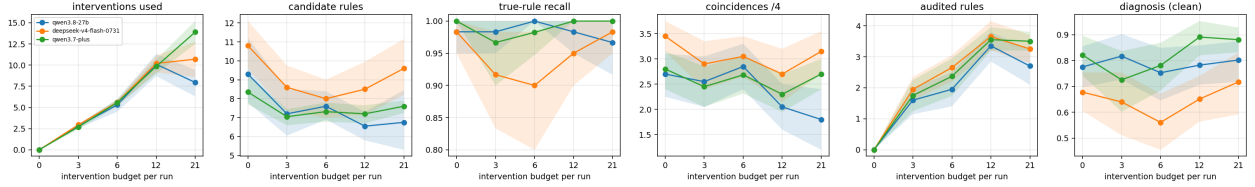


Figure 2: Intervention budget ladder. Belief-level spurious adoption is flat; audited rules grow with budget; budget stops binding above 12.

measure and explains the main effect: the gate helps because the audited set contains something, and the audited set contains something because budget was spent.

5.6 Budget is not a strictness dial

Raising the budget from 0 to 21 leaves spurious adoption essentially unchanged (only one adjacent step survives correction, and the next step reverses) and leaves recall unchanged. What grows is the number of *audited* rules (0 $\rightarrow$ 3.35–3.65; Figure 2). Giving an agent more opportunities to verify does not stop it believing coincidences; it only increases the stock of evidence available to the action layer. This reinforces the mechanism reading of §5.4.

5.7 T3 refuted: strict agents do not stop exploring

Across six strictness levels and three models, intervention usage stays at the budget ceiling (4.80–6.00, 5.85–6.00, 6.00), and exploration breadth does not collapse. Every adjacent contrast on intervention count has $p_{\text{Holm}} > 0.29$.

A discovery cliff does exist, but not where T3 predicted. Per-level recall loss is essentially zero from L0 to L2; it appears at L3 $\rightarrow$ L4 (-0.091 , $p_{\text{Holm}} = .088$, marginal) and L4 $\rightarrow$ L5 (-0.152 , $p_{\text{Holm}} = .048$). The cliff is located where the requirement changes from subjective confidence to an objective evidential threshold, not at the far end of a verbal strictness scale. The four purely verbal levels L0–L3 are nearly free: candidate rules fall from 28.9 to 5.95 in one model while recall drops only 0.05. Prompt-level strictness mostly removes redundant candidates, not discovery capability (Figure 3).

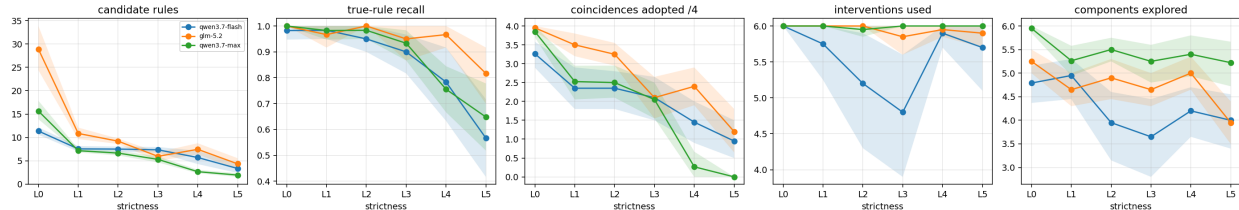


Figure 3: Strictness ladder. Interventions used (fourth panel) is flat, refuting the exploration-collapse hypothesis; the recall cliff appears only at L4–L5.

5.8 Robust to the environment, sensitive to the model

On a previously unused model, across all five environment variants including the one equivalent to the main experiments, the pooled gate effect is $+0.006$ $[-0.077, +0.088]$, $n = 40$, with no variant significant. Spurious adoption tracked the environment as designed (1.6/4 under WEAK to 3.9/4 under STRONG), so the manipulation worked. The result is robust to environment parameters and not robust to model choice. Under the moderation structure of §5.4 this non-replication is expected: that model’s baseline was 0.78–0.90, inside the zero-benefit region.

6 Discussion

Caution is not free, and not uniformly expensive. Strictness reliably suppresses superstition. It costs recall, and in some models a great deal of it, but the cost is not universal, and verbal strictness is cheap over a wide range. What is expensive is the step from “be confident” to “have interventional evidence”.

What an action gate actually does. It does not make the agent smarter. In the conditions where it helps most, the belief set still contains three spurious rules of four; the gate severs the path from those rules to action. Two conditions follow, and both are engineering-relevant: the gate requires that the system can already verify (at zero budget it costs 0.315 accuracy), and it requires that the system is not already performing well (benefit $+0.31$ below baseline 0.60 versus $+0.01$ at 0.80–0.95).

Tension with capability-scaling results. The Hypothesis Evolution Protocol (HEP, 2026) reports that stronger base models exploit an auditable hypothesis protocol *more* fully, whereas we find the gate’s accuracy benefit *shrinks* as the ungated baseline rises. The measured quantities differ—protocol utilisation versus downstream accuracy gain—and we cannot resolve the tension with our data, but the two should not be conflated. It is possible for a strong model to use a protocol more thoroughly while gaining less from it.

A negative result worth stating. Giving an agent more chances to verify does not make it stop believing coincidences. Budget changes the evidence stock, not the belief set. This suggests the belief layer and the evidence layer are only weakly coupled in current LLM agents, which is precisely why an action-side gate, rather than a belief-side intervention, is the effective lever.

7 Limitations

1. Chain-of-thought was disabled throughout. Whether these conclusions hold with reasoning enabled is untested.
2. The model set differs between experiments because of quota availability, and some cells are small (minimum 6 paired observations). Results are not pooled across experiments.
3. No single model survives Holm correction for the T2 main effect; it is detectable only in the pooled sample.
4. The split-half is a fixed parity split rather than a randomised one.
5. The budget-by-gate interaction was measured under the earlier 8-case protocol and, by analogy with §5.4, its magnitude is probably overestimated.
6. Five environment variants were tested but all belong to one generative family; non-binary symptoms, temporal dependence and multi-cause interactions are untested.
7. Models used are mid-sized and inexpensive; frontier models are untested.

8 Reproducibility

Code, data and per-run records are available at <https://github.com/aegisafety/fault-world>. Every run is checkpointed with its complete conversation, executed interventions and diagnostic answers, so all reported numbers can be recomputed from raw records without API access. The release contains the environment (`fault_world_v2.py`, `fault_world_v3.py`), the step-wise resumable experiment driver (`run_2x2.py`), the audit and oracle code (`audit_gate.py`), and the analysis scripts (`analyze_2x2.py`, `stats.py`, `holm.py`, `t3_curve.py`, `s2_curve.py`, `bg_interaction.py`, `e4_robustness.py`, `t2_replication.py`). In total 1,330 runs completed across 18 experiment cells.

A Hypotheses of ours that the data refuted

Original hypothesis	Outcome
Over-strict agents stop exploring (T3)	Refuted: intervention usage flat across six levels
A single model-independent FDR–recall frontier exists	Not supported: the cost is model-dependent
Prompt-level strictness is the primary independent variable	Downgraded: L0–L3 nearly free
Self-reported VERIFIED labels can be trusted	Refuted: must be audited by the harness
Strictness induces Goodharting of status labels	Not replicated: single-model observation
Intervention budget is a form of strictness	Refuted: it only grows the evidence stock
The action gate extends the frontier unconditionally	Conditional: harmful at zero budget
The gate’s benefit is general	Conditional: strongly moderated by the un-gated baseline
Strictness reduces intervention count	Withdrawn after Holm correction ($p = .108$)
Evidential requirements increase exploration	Withdrawn after Holm correction ($p = .294$)

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